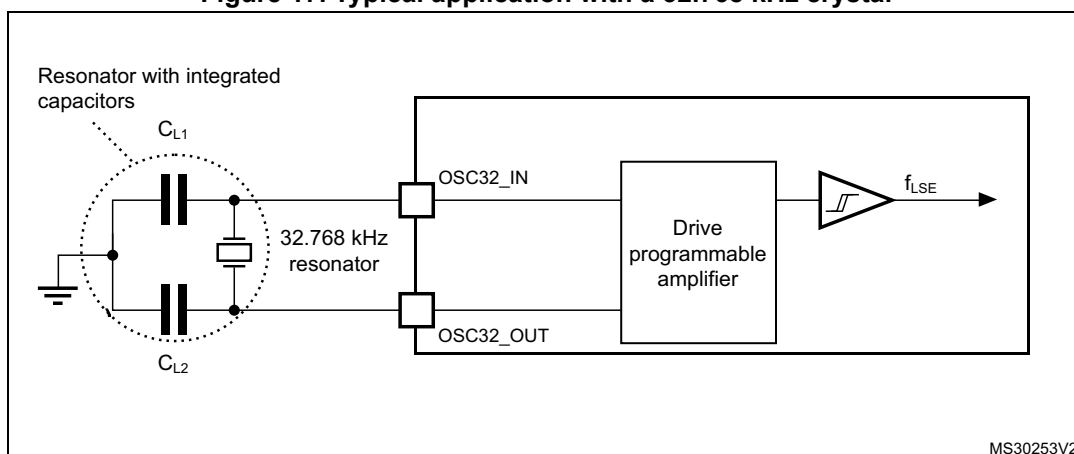


Figure 17. Typical application with a 32.768 kHz crystal



Note: An external resistor is not required between OSC32_IN and OSC32_OUT and it is forbidden to add one.

5.3.8 Internal clock source characteristics

The parameters given in [Table 41](#) are derived from tests performed under ambient temperature and supply voltage conditions summarized in [Table 24: General operating conditions](#). The provided curves are characterization results, not tested in production.

High-speed internal (HSI48) RC oscillator

Table 41. HSI48 oscillator characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{HSI48}	HSI48 Frequency	$V_{\text{DD}}=3.0\text{ V}$, $T_{\text{A}}=30\text{ }^{\circ}\text{C}$	47.92	-	48.40	MHz
$\Delta_{\text{Temp(HSI)}}^{(1)}$	HSI48 oscillator frequency drift over temperature and V_{DD} full voltage range	$T_{\text{A}}= 0\text{ to }85\text{ }^{\circ}\text{C}$	-1	-	1	%
		$T_{\text{A}}= -40\text{ to }125\text{ }^{\circ}\text{C}$	-2.5	-	2	%
$\text{TRIM}^{(1)}$	HSI48 oscillator frequency user trimming step	From code 127 to 128	-8	-6	-4	%
		From code 63 to 64 From code 191 to 192	-5.8	-3.8	-1.8	
		For all other code increments	0.2	0.3	0.4	
$D_{\text{HSI48}}^{(2)}$	Duty cycle	-	45	-	55	%
$t_{\text{su(HSI48)}}^{(2)}$	HSI48 oscillator start-up time	-	-	1.4	1.8	μs
$t_{\text{stab(HSI48)}}^{(2)}$	HSI48 oscillator stabilization time	at 1% of target frequency	-	1.5	3.6	μs
$I_{\text{DD(HSI48)}}^{(1)}$	HSI48 oscillator power consumption	-	-	525	570	μA

1. Based on characterization results, not tested in production

2. Specified by design – Not tested in production.